The Invisible Forests That Keep Earth Alive

Introduction

What are phytoplankton?

When people think about plants that produce oxygen, they usually imagine forests like the Amazon rainforest. However, some of the most important "plants" on Earth are actually **microscopic organisms floating in the ocean**.

Phytoplankton are tiny photosynthetic organisms that drift with ocean currents. Although each individual cell is invisible to the naked eye, together they form enormous populations that can even be detected by satellites from hundreds of kilometers above Earth.

Like plants on land, phytoplankton use sunlight, water, and carbon dioxide to produce food through **photosynthesis**.

Photosynthesis

$$6CO_2 + 6H_2O + sunlight \rightarrow C_6H_{12}O_6 + 6O_2$$

This process not only provides energy for phytoplankton but also releases oxygen into the atmosphere.

Many people assume that most of the oxygen in that breath came from forests. While forests are incredibly important, approximately half of the oxygen produced on Earth each year comes from marine phytoplankton.

This is why scientists often call phytoplankton the **"lungs of the ocean."** Although tropical rainforests are sometimes referred to as the "lungs of the Earth," they consume much of the oxygen they produce through respiration and decomposition. In contrast, phytoplankton continuously generate enormous amounts of oxygen while also supporting the marine food web.

Although phytoplankton are microscopic—many measuring less than **50 micrometers**, or about half the width of a human hair—they form the biological foundation of nearly every marine ecosystem.

Sunlight → Phytoplankton → Zooplankton → Small Fish → Large Fish → Sharks • Tuna • Seabirds • Whales • Humans

If phytoplankton populations decline, the effects cascade upward through every trophic level. Fewer phytoplankton mean less food for zooplankton, leading to reduced fish populations and ultimately affecting marine mammals, seabirds, commercial fisheries, and the millions of people who rely on seafood for food security and livelihoods. NOAA describes phytoplankton as the **base of most marine food webs,** making them essential to ocean ecosystem functioning.

The Ocean's Hidden Carbon Capture System

Phytoplankton do much more than produce oxygen.

They also remove carbon dioxide from the atmosphere.

During photosynthesis, phytoplankton absorb CO_2. When they die, many sink to the ocean floor, carrying that carbon with them. This process is known as the **biological carbon pump**.

This natural system helps slow the accumulation of greenhouse gases in the atmosphere and plays a critical role in regulating Earth's climate.

Scientists estimate that the ocean absorbs roughly **30% of human-produced carbon dioxide emissions** and about **90% of the excess heat** trapped by greenhouse gases, making healthy ocean ecosystems vital in moderating climate change.

Our Research

Boston sits on **Massachusetts Bay**, which is part of the **Gulf of Maine**, one of the world's most productive temperate marine ecosystems.

Several factors make this region ideal for studying phytoplankton:

- Cold, nutrient-rich water from the Gulf of Maine
- Strong seasonal changes in temperature
- Frequent winter storms and spring mixing
- Large freshwater inputs from rivers
- Extensive fisheries and diverse marine wildlife

These conditions create large seasonal phytoplankton blooms that are easily observed by satellites and make Massachusetts Bay a natural laboratory for studying ocean ecosystems.

Our research studies four major groups of phytoplankton. Together, these four groups account for much of the phytoplankton community in the Gulf of Maine and respond differently to changing environmental conditions.

Group	Why they're important

Diatoms	Build glass-like silica shells. Responsible for many of the large spring blooms and play a major role in transporting carbon to the deep ocean.
Dinoflagellates	Can swim using tiny flagella. Some species create spectacular bioluminescence, while others cause harmful algal blooms (red tides).
Green algae	Photosynthetic algae related to the ancestors of land plants. Common in nutrient-rich coastal waters.
Prymnesiophytes	Includes coccolithophores, which build calcium carbonate plates and are important for Earth's carbon cycle.

Although they all belong to the phytoplankton community, each group responds differently to temperature, nutrients, sunlight, and ocean mixing, which is why our project compares their seasonal patterns.

The Regions Chosen

We chose 4 regions to study, from 2016-2025.

Region 1: Offshore of Massachusetts Bay / Stellwagen Bank

- Underwater plateau located at the mouth of Massachusetts Bay, between Cape Ann and Cape Cod
- Semi-enclosed basin, driven by deep-water nutrient upwelling, fed by Gulf of Maine Coastal Current (GMCC)
- Strong tidal mixing over the shallow bank meets stratified water in the deeper bay, creating a persistent tidal front
- Front continuously resupplies nutrients while concentrating phytoplankton and grazers
- Stable, elevated productivity

Region 2: East of Cape Cod

- Water exiting Massachusetts Bay and the GMCC wraps around the outer Cape here and starts mixing with open shelf water (a transition zone)
- Shaped more by wind and current variability than fixed tidal mixing, as Gulf of Maine water mixes with open shelf water
- Productivity is patchier and more episodic, tracking storm events and shifts in the current boundary rather than a stable front

Region 3: South of Cape Cod / near Nantucket Shoals

- Intense tidal mixing keeps the shoals well-mixed year-round, setting up a persistent front where they meet stratified water offshore. The shoals themselves see continuous but often light-limited production, while the front acts as a reliable productivity hotspot.

Region 4: Further offshore south of Nantucket

- Deeper water here allows classic seasonal stratification
- Drives a textbook bloom pattern (strong spring bloom, summer subsurface chlorophyll max, and a fall bloom when mixing resumes)

The Data Collection Process

1. Landsat 8-9 images were pulled from USGS Earth Explorer for each of the four regions, then processed through SeaDAS.
2. Each pixel represents light reflected off the ocean at blue, green, and red wavelengths. Chlorophyll absorbs and reflects light differently based on concentration, so the Blue-Green Ratio algorithm converts these reflectance values into a chlorophyll-a estimate. Pixels blocked by clouds can't have chlorophyll measured underneath, so they're flagged as NaN using a separate cloud brightness value. This is the "No data (cloud cover)" category in the bars.

$$\frac{R_{443}}{R_{560}}$$

3. SeaDAS also applies the Hirata algorithm to split chlorophyll into functional groups (diatoms, dinoflagellates, green algae, prymnesiophytes).
4. Stripe correction (MATLAB) removes sensor artifacts, refined in Photoshop via masking, so pixel values are not skewed by processing errors.
5. Each region is defined by fixed lat/long boundaries. Chlorophyll values within those boundaries are exported as CSVs for every date, then aggregated into monthly bars.
6. CSVs across 2016 to 2024 were compiled into the bar graphs, heatmaps, and trendlines with outliers (red stars) flagged where concentration spiked well above the region's typical range, allowing comparison against known marine heatwave events.

Data Automation process

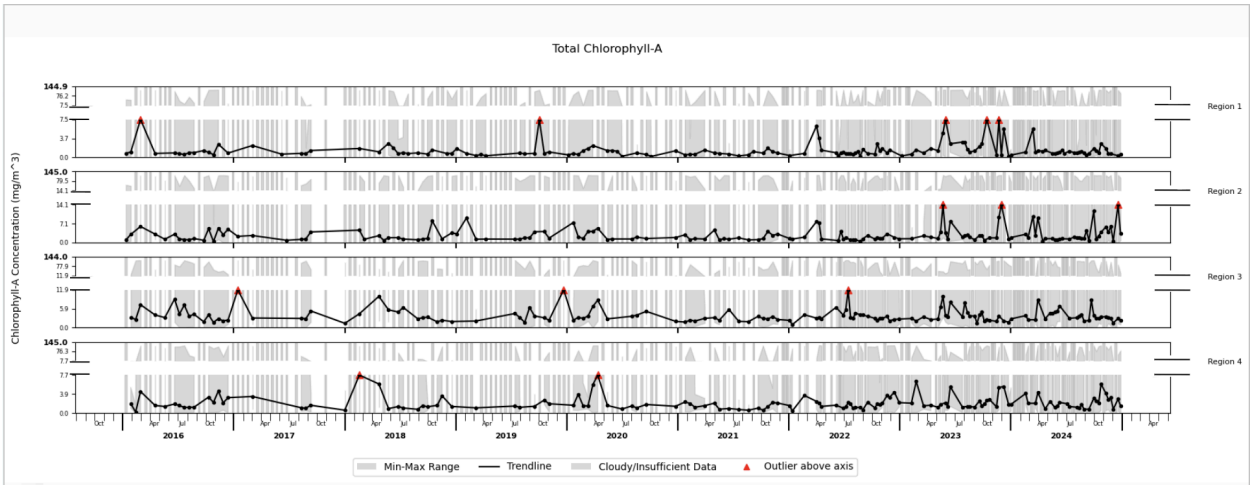
Previously, the entire process of creating a visualization of one date manually took multiple hours and intensive care. This causes the process to complete each date very gradually, and other methods would be preferred to speed up the process. As a result, we have created an automation pipeline that completes the destriping much faster than if a person tried it by hand.

The automation pipeline consists of multiple Python files and packages. Firstly, a Python script inputs a date and retrieves the satellite Landsat data by placing an order on the ESPA website. The script automatically downloads the data and creates a CSV that describes the phytoplankton concentration in four regions of interest. Afterwards, it processes the raw satellite data to make initial RBG images of the different types of phytoplankton through L2gen. Finally, the post-processing pipeline rotates, interpolates, and destripes those images, which mimics the manual Photoshop process that takes hours.

Through the pipeline, we will be able to drastically minimize the time it takes to make visualizations. We are currently working on further improving the pipeline by perfecting the destriping algorithm through various methods. By implementing these improvements, we hope to create the most accurate visualizations to date.

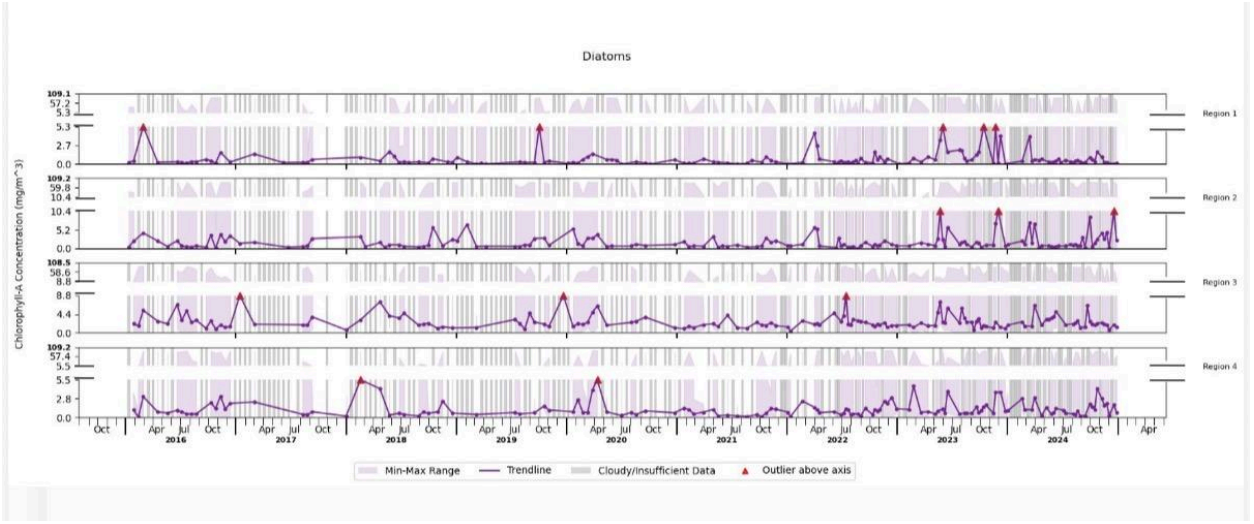
Trendlines

Total Chlorophyll-A Graph



The total chlorophyll-A graph looks very similar to the **diatoms** graph. Suggests diatoms may be the main driver of total chlorophyll-A peaks (which aligns with the high concentration of diatoms as shown in the bar graph)

Diatoms



Diatom concentration peaks:

- Region 1: January 2016, October 2019, June, July & October 2023
- Region 2: June & November 2023, December 2024
- Region 3: January 2017, December 2019, July 2022
- Region 4: March 2018, June 2020

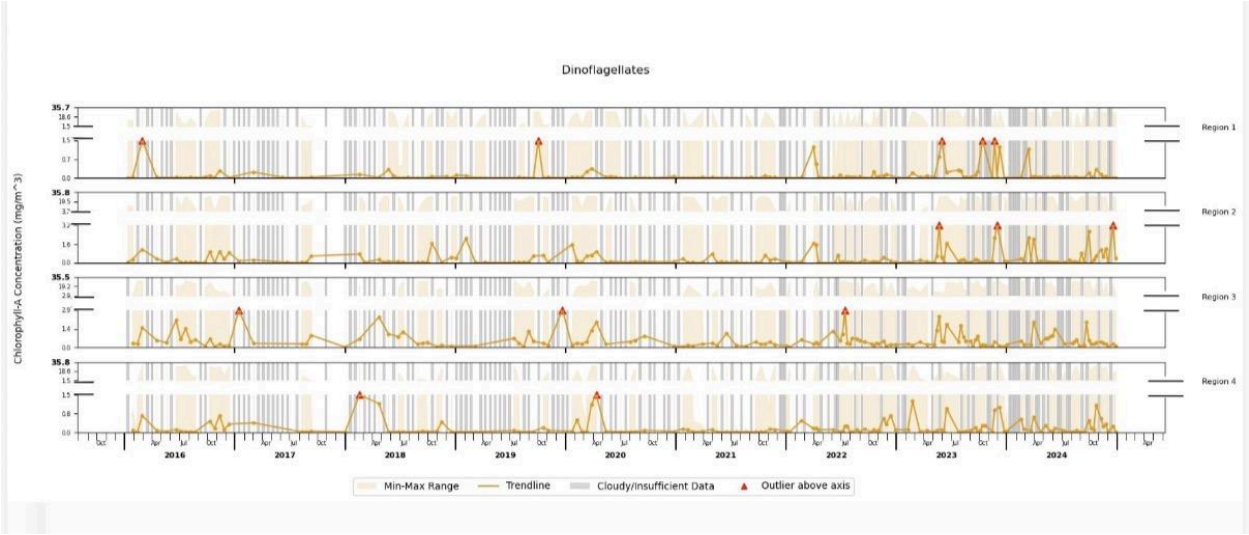
Traditionally: Diatom populations typically reach their maximum in early spring (roughly

March–May), then taper off through the summer as the water column stratifies. Production picks back up again in fall, once mixing returns and breaks down that stratification.

In our data: 8 of 11 outliers peak in months the classic model predicts low diatoms. Suggests these regions don't follow typical temperate stratification/light-limitation dynamics, likely driven by local upwelling, mixing, or non-temperate conditions instead

Edwards, M., Beaugrand, G., Kléparski, L. *et al.* Climate variability and multi-decadal diatom abundance in the Northeast Atlantic. *Commun Earth Environ* **3**, 162 (2022). https://doi.org/10.1038/s43247-022-00492-9

Dinoflagellates

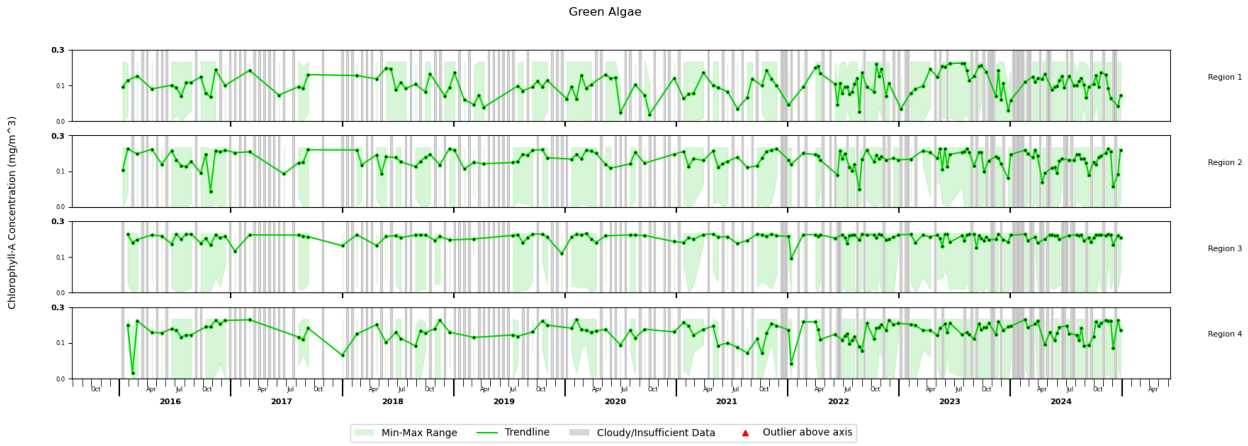


Dinoflagellate concentration peaks:

- Region 1: February 2016, October 2019, June & September & October 2023
- Region 2: June & November 2023, December 2024
- Region 3: February 2017, November 2019, July 2022
- Region 4: February 2018, April 2020

Dinoflagellates traditionally bloom opposite diatoms, favoring calm, warm, stratified water from late spring through early autumn (May to October). Their motility lets them swim toward light and nutrients instead of relying on turbulence. In this data, only about half the peaks (5-6 of 12) fit that window. The rest cluster in winter, with several regions showing diatom and dinoflagellate peaks overlapping in winter too. This suggests local drivers matter more here than the traditional seasonal cycle. Winter peaks fit Massachusetts coastal dynamics well. Winter storms mix nutrients up from depth, and freshwater runo from snowmelt and rain restratifies the surface enough to support blooms even in low light. Nutrient loading from runo and storm or tidal mixing likely matter more here than light and stratification alone. Normally diatoms bloom first and decline before dinoflagellates take over. In this data, both groups peak at similar times in the same regions. This points to a different driver, likely storm-driven nutrient pulses and resuspension of dinoflagellate cysts, rather than the classic succession pattern.

Green Algae

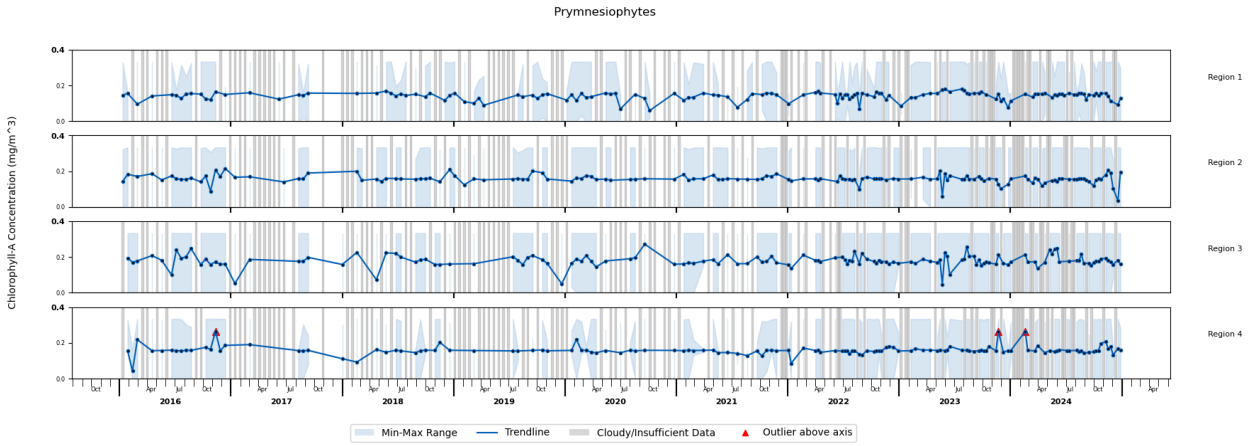


- Region 1 has the lowest and most volatile chlorophyll-a of the four regions. Tidal mixing and Gulf of Maine circulation prevent sustained blooms.
- Region 2 is much more stable and moderate. Typical of open Atlantic shelf water with less extreme swings.
- Region 3 has the highest sustained concentrations, driven by shallow depth and strong tidal nutrient resuspension
- Region 4 is volatile, consistent with a transition zone between shoal productivity and open shelf water. Shows the clearest anomaly, a dramatic dip in 2016

For the seasonal pattern of green algae concentration, there is typically a strong bloom in the late spring as the water warms and daily sunlight increases. This is followed by a trough in the late summer, as they become limited by nutrients, there is high heat and low nitrogen levels. Then during the summer, as daily sunlight becomes limited, visible concentrations disappear from the surface of the water. Green algae are also sensitive to their geographic location and much higher near the shoreline, estuaries, and river mouths.

Region 2 and 3 follow this pattern better with relatively consistent oscillations without extreme troughs, while Regions 1 and 4 show sharper, more irregular crashes that are more likely due to storms.

Prymnesiophytes



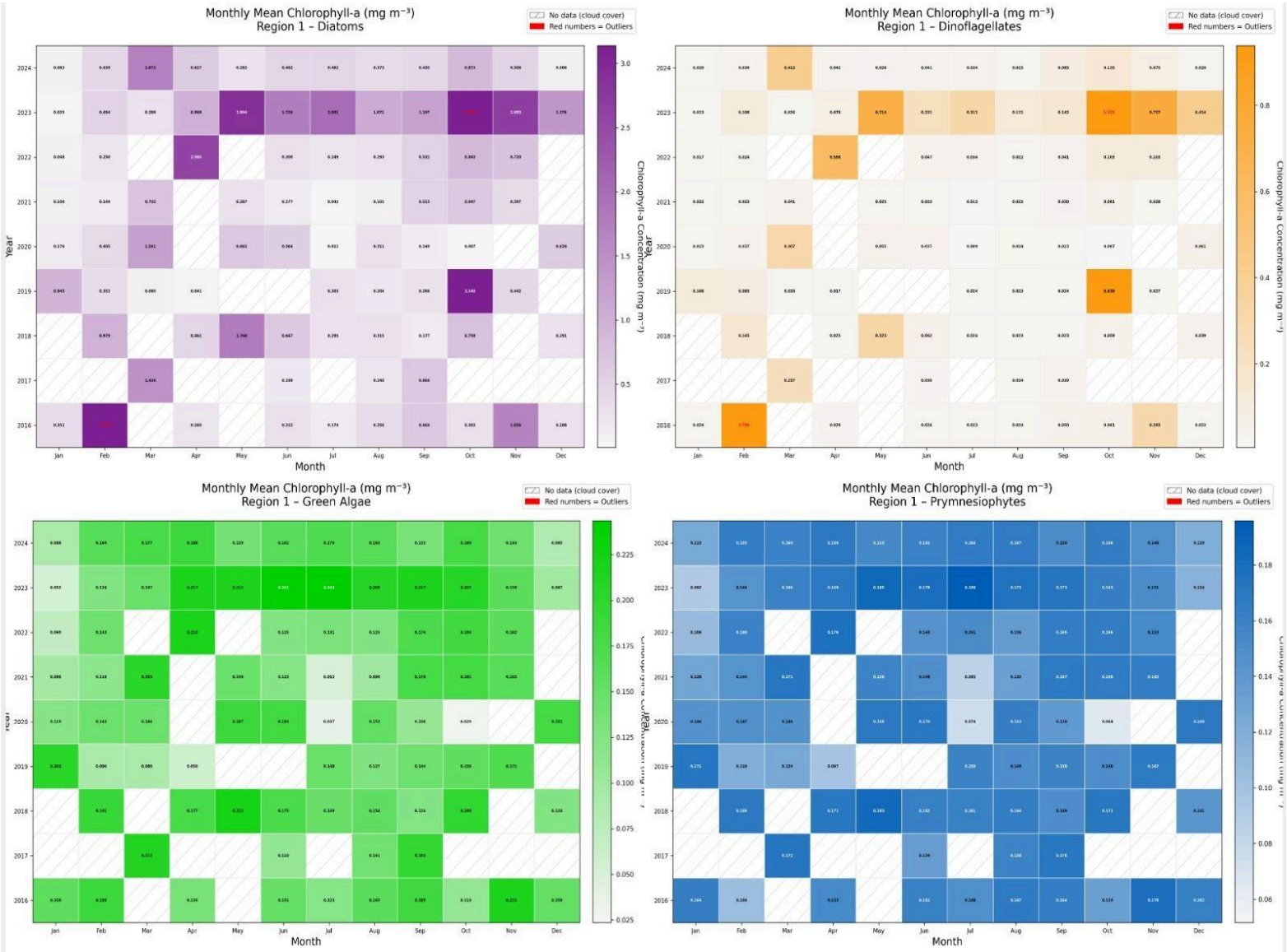
- Region 1 has the lowest concentrations and is more stable. Consistent with a tidally mixed, less stratified environment.
- Region 2 is relatively moderate and stable as well with few crashes
- Region 3 is more volatile with dips in early 2017, 2018, and late 2019
- Region 4 is the most volatile with the only flagged outliers in late 2016, late 2023, and early 2024

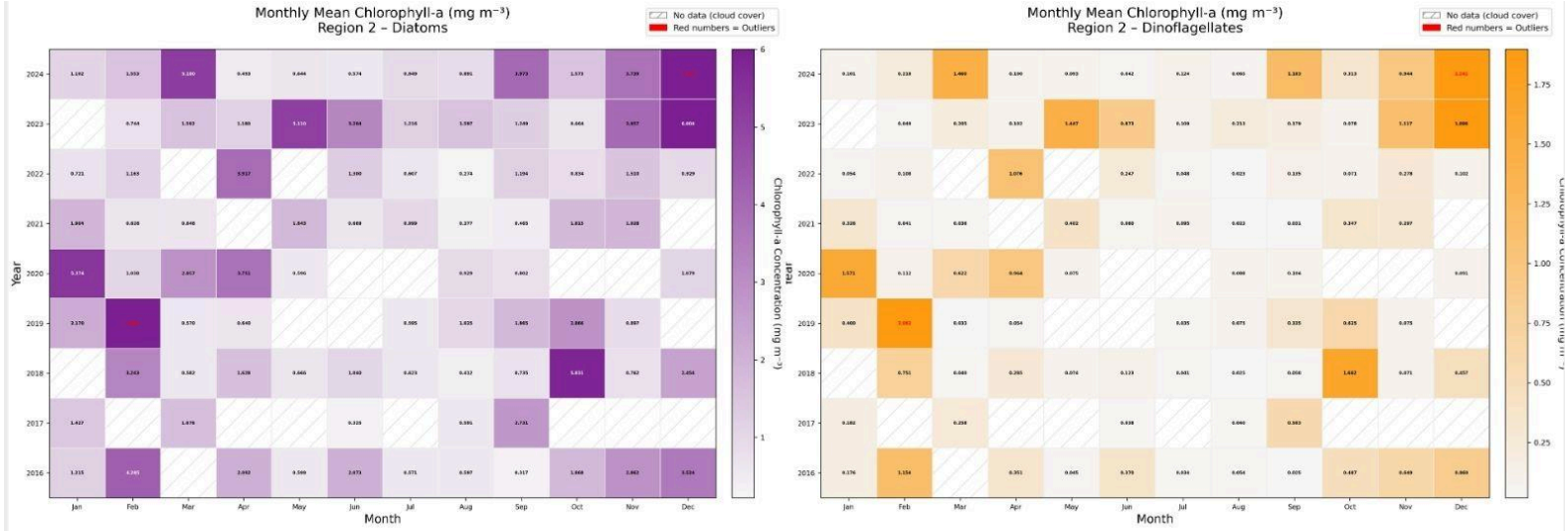
Prymnesiophytes typically peak in late spring to summer, since diatoms deplete silica and the water stratifies, and often have a longer, lower-amplitude bloom than spring diatom blooms. None of the regions really follow this pattern, and might be affected more strongly by storms or gaps in data due to clouds.

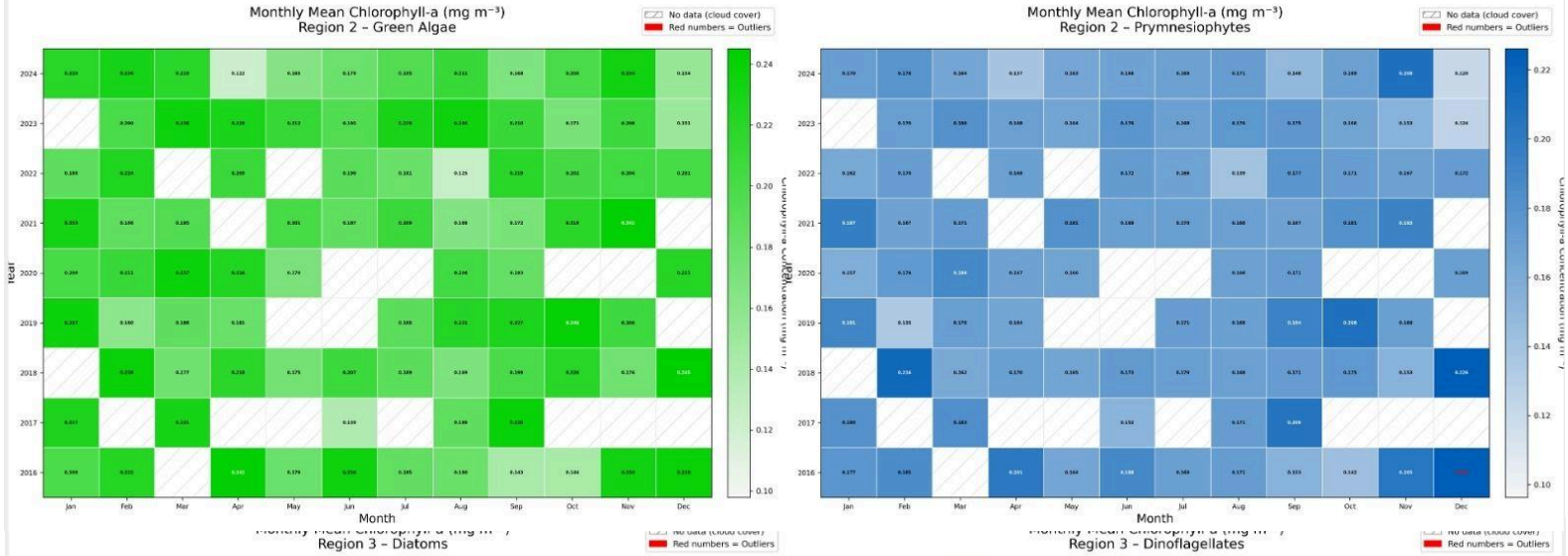
Heatmaps

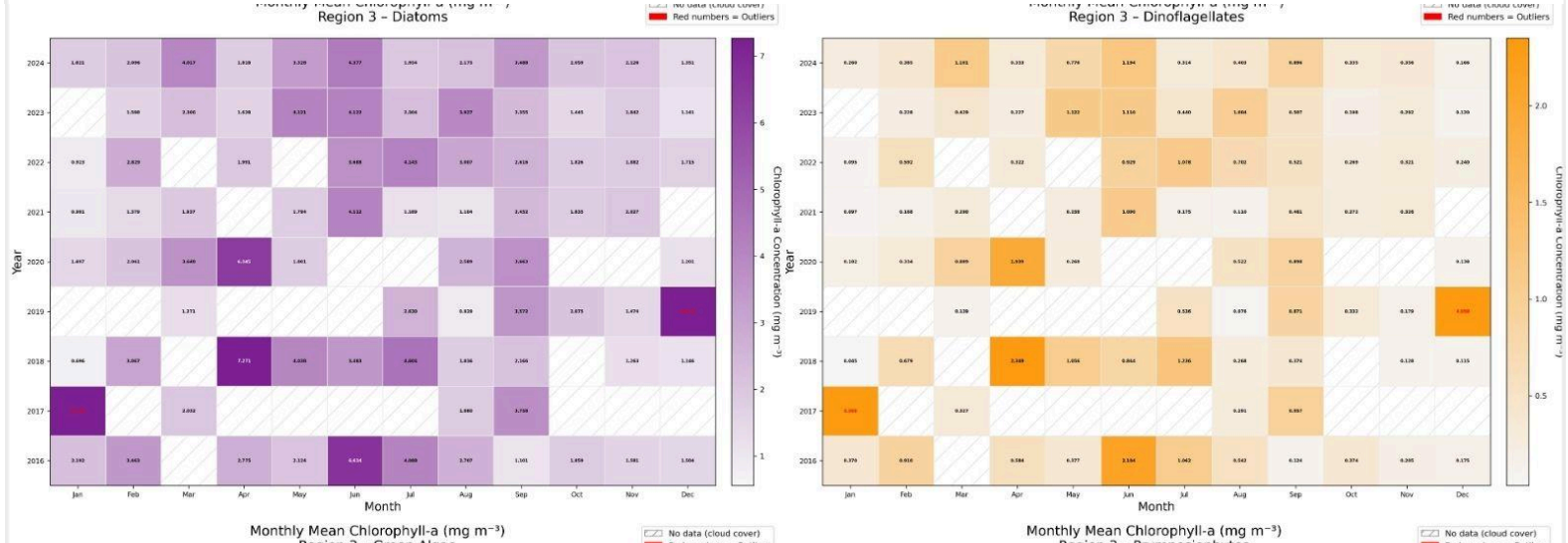
The heatmaps below show the monthly average chlorophyll-a concentration for region 1 to 4, from 2016-2024. The rows represent the years, and columns represent the months, and each cell shows the average chlorophyll-a concentration of a particular phytoplankton (green algae, diatoms, dinoflagellates, prymnesiophytes). Reading across a row allows you to observe how chlorophyll-a changes throughout a single year (seasonal patterns), while reading down a column to compare the same month across different years (year-over-year trends).

- **Color**: Represents concentration, the darker the color, the higher the chlorophyll-a level that month.
- **Clusters of dark cells**: Represent the bloom outliers
- **Hatching**: Indicates missing data (usually due to cloud cover)
- **Red numbers**: Mark statistical outliers.

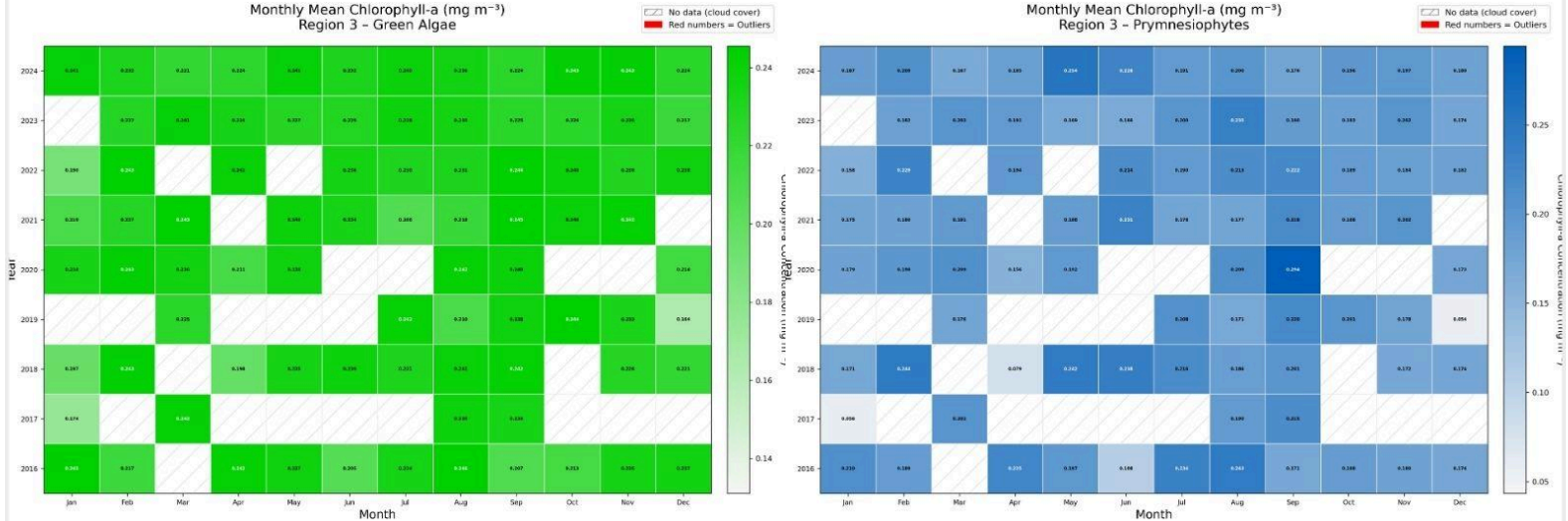



Monthly Mean Chlorophyll-a (mg m⁻³)
Region 2 – Diatoms
No data (cloud cover)
Red numbers = Outliers
Year
Month
Chlorophyll-a Concentration (mg m⁻³)


Monthly Mean Chlorophyll-a (mg m⁻³)
Region 2 – Dinoflagellates
No data (cloud cover)
Red numbers = Outliers
Year
Month
Chlorophyll-a Concentration (mg m⁻³)


Monthly Mean Chlorophyll-a (mg m⁻³)
Region 2 – Green Algae
No data (cloud cover)
Red numbers = Outliers
Year
Month
Chlorophyll-a Concentration (mg m⁻³)

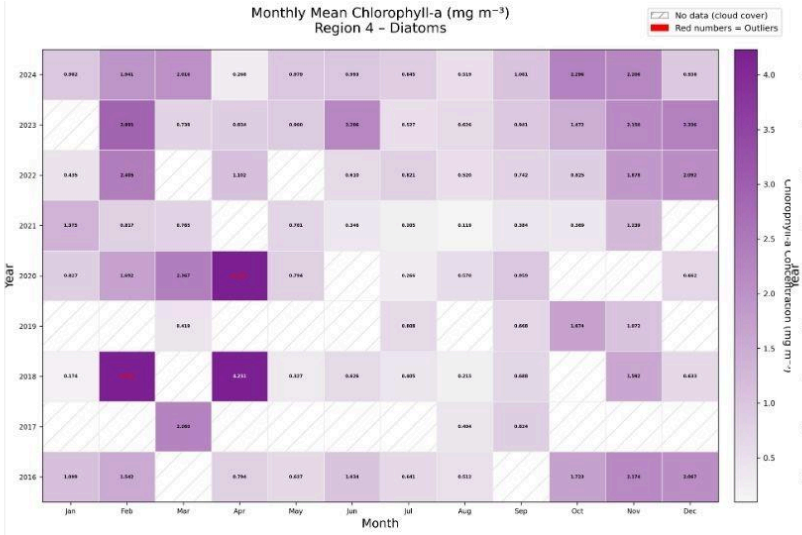
Monthly Mean Chlorophyll-a (mg m⁻³)
Region 2 – Prymnesiophytes
No data (cloud cover)
Red numbers = Outliers
Year
Month
Chlorophyll-a Concentration (mg m⁻³)

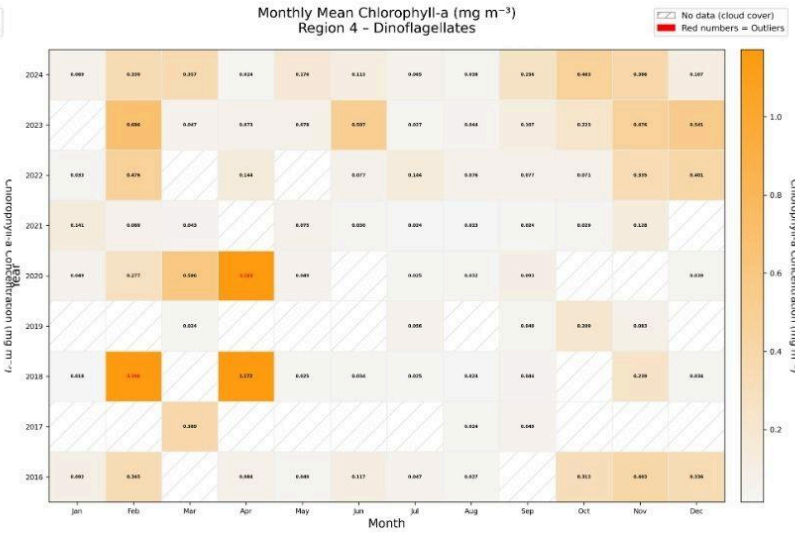

Monthly Mean Chlorophyll-a (mg m⁻³)
Region 3 – Diatoms
No data (cloud cover)
Red numbers = Outliers
Year
Month
Chlorophyll-a Concentration (mg m⁻³)

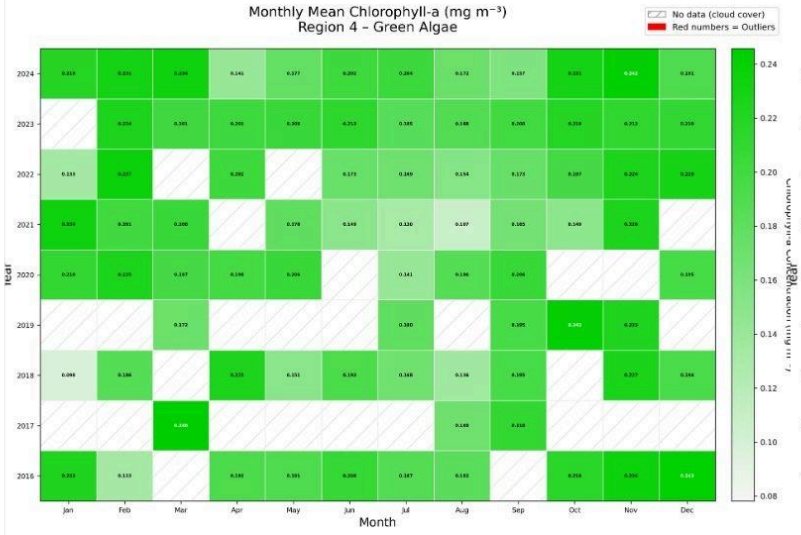
Monthly Mean Chlorophyll-a (mg m⁻³)
Region 3 – Dinoflagellates
No data (cloud cover)
Red numbers = Outliers
Year
Month
Chlorophyll-a Concentration (mg m⁻³)

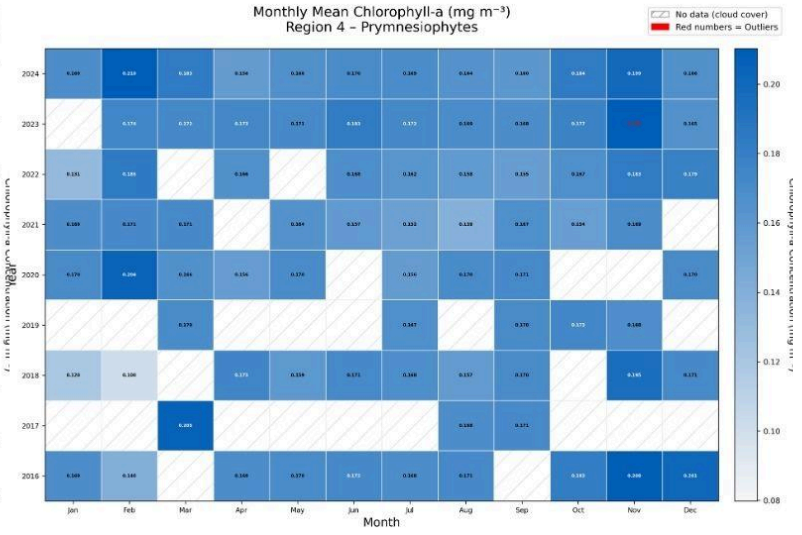
Monthly Mean Chlorophyll-a (mg m⁻³)
Region 3 – Green Algae
No data (cloud cover)
Red numbers = Outliers
Year
Month
Chlorophyll-a Concentration (mg m⁻³)

Monthly Mean Chlorophyll-a (mg m⁻³)
Region 3 – Prymnesiophytes
No data (cloud cover)
Red numbers = Outliers
Year
Month
Chlorophyll-a Concentration (mg m⁻³)









External Factors

<u>January 2017</u>

1. Northeast average temperature: 6.8°F above normal 9th warmest January since 1895
   Massachusetts itself ranked among its 10 warmest Januaries on record.
   - warmer water than typical
   - delayed winter cooling
   - different stratification patterns

2. The Northeast received 118% of normal precipitation during January.
   This could increase river discharge, coastal nutrient input, and estuarine flushing

3. NOAA documents several major storm systems:
    - January 4–8 Large lake-effect snow event.
    - January 7–8
    - January 23–25

Winter storms create stronger vertical mixing, deeper mixed layers, sediment resuspension and nutrient entrainment
For coastal regions like Nantucket Shoals Georges Bank winter mixing is one of the classic mechanisms that replenishes nutrients before spring blooms.

February 2018

1. Unprecedented mid-month heatwave - February averaged 6.4 degrees above average
   Year tied for the warmest February on record with 1925, averaging 38.1 degrees
2. Between the end of February and early March 2018, a series of massive storm systems (nor'easters) brought strong winds, heavy snow, and tremendous coastal flooding to communities from the Mid-Atlantic to northern Maine
   Storms at this time are not unusual for Massachusetts, but their strength and close timing were
3. Region specific: Smith et al investigated Phaeocystis pouchetii (cold-water marine alga famous for forming massive blooms), and reported a large bloom off the coast of New England during early spring 2018, centered on Nantucket Shoals
   Attribute it to a relaxation of physical conditions after a period of storm-driven mixing. Once wind stress eased, the water column could restratify while nutrients that had been mixed up during the stormy period were still sitting in the surface layer

February 2019

1. The Northeast was about 1.2°F warmer than normal, and the overall winter was also warmer than average.

2. February precipitation = 123% of normal
   Winter precipitation = 129% of normal
   Massachusetts = 114% of normal winter precipitation

More precipitation can lead to greater river discharge, increased nutrient transport, enhanced coastal productivity.

3. Storms

can deepen the mixed layer, transport nutrients upward, break down short-term stratification, and redistribute phytoplankton.
Fits diatoms particularly well because they respond rapidly to newly available nutrients.

December 2019

1. NOAA reports that the Northeast averaged 1.7°F above normal, and Massachusetts itself finished the month slightly above normal. Warmer temperatures alone do not cause blooms, but they may influence water column stability and phytoplankton growth rates

2. Massachusetts experienced 170% of its normal December precipitation, making it the

7th wettest December on record.
- more rainfall
- more runoff
- more freshwater entering coastal waters
- can transport nutrients into coastal ecosystems

3. NOAA documents several major storm systems:
   - Dec 1–3
   - Dec 13–14
   - Dec 16–17
   - Dec 29–31

increase wind-driven mixing alter coastal circulation increase freshwater runoff

<u>January 2020</u>

- Mean diatoms ≈ 5.37 mg m^{-3}
- Top 10% diatoms ≈ 30.94 mg m^{-3}

One of the largest localized blooms in the entire dataset.

1. Massachusetts actually experienced its 10th driest January on record.

The largest bloom did not occur during an especially wet month-> river runoff wasn't the dominant driver (?)

2. Storm on January 25 NOAA

If January satellite observations occurred after this event, storm-induced mixing (?)

East of Cape Cod= influenced much more by shelf circulation, tidal mixing, exchange with the Gulf of Maine, than by river discharge.
January 2020 was relatively dry, but still observed an extremely large diatom bloom
The bloom may have been maintained by physical oceanographic processes already present on the shelf. East Cape Cod blooms may be controlled by ocean circulation and seasonal mixing rather than by rainfall-driven nutrient pulses.

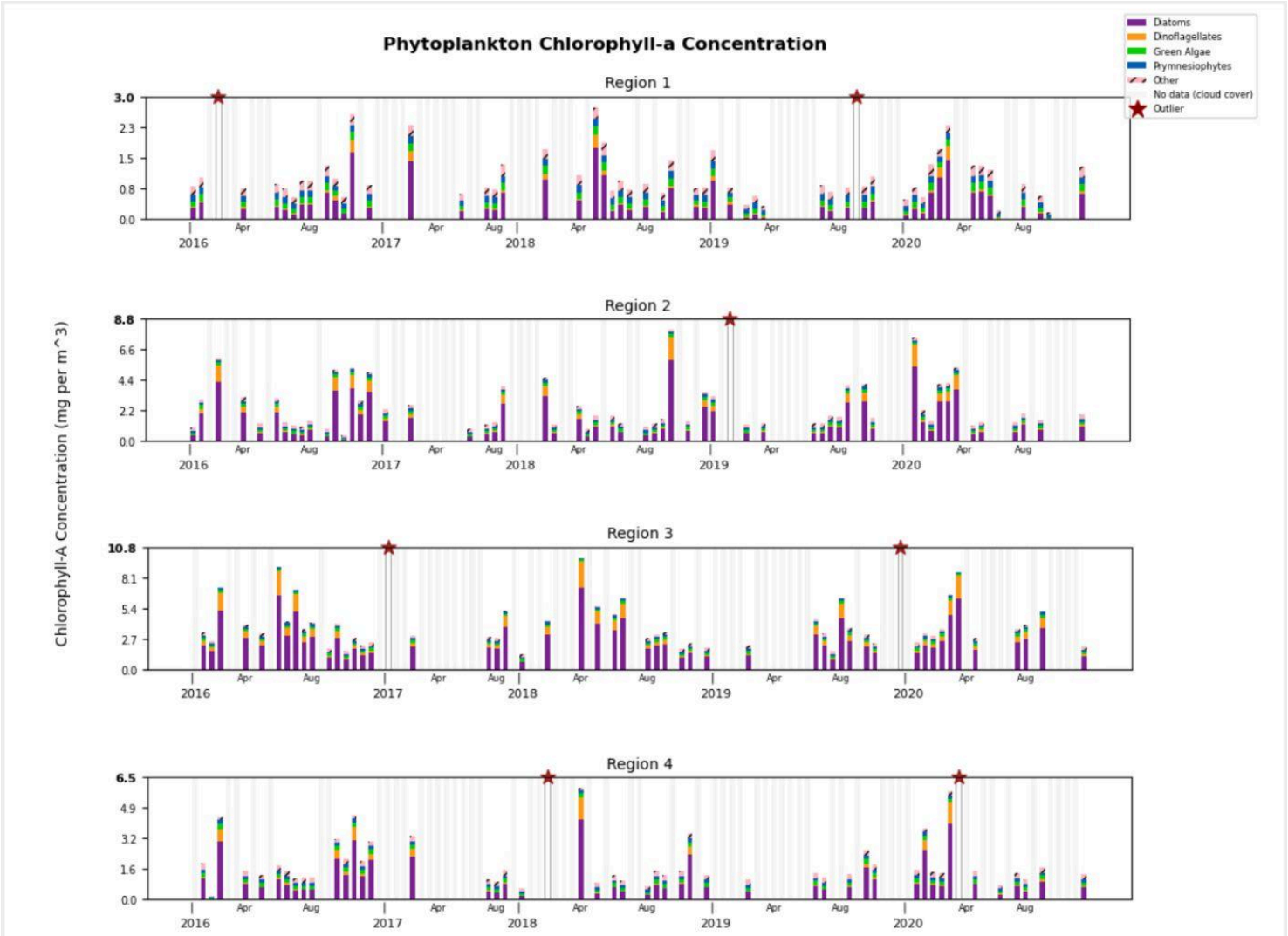
<u>April 2020</u>

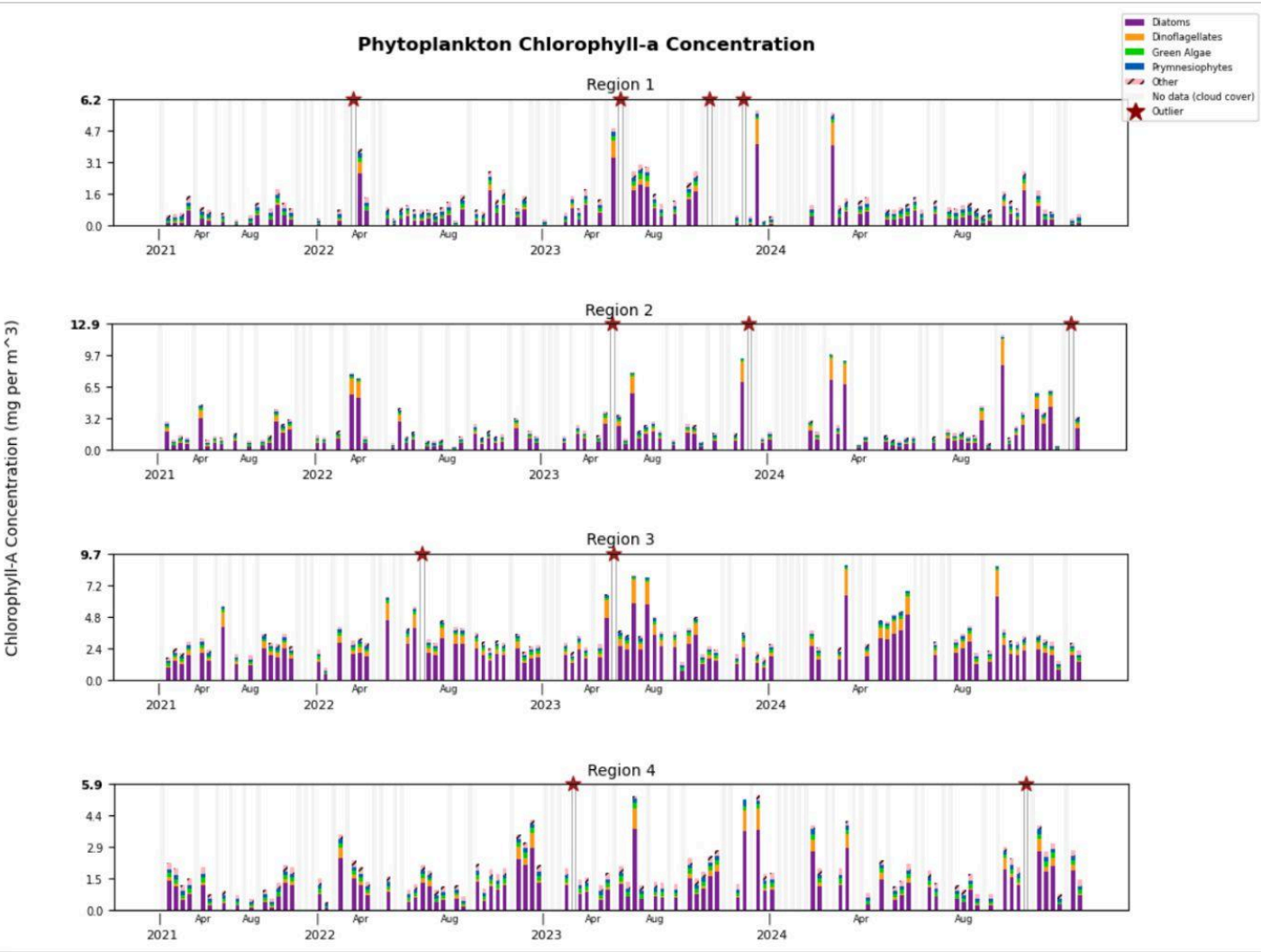
1. Unusually cold, wet & stormy
   Late-season snowfall on April 18 and a severe nor'easter on April 13
   April 3 - powerful coastal storm hit the Cape Cod/Nantucket/Martha's Vineyard area directly, brought coastal flooding on Nantucket
2. Already aligns with seasonality
   Climatological peak of the spring phytoplankton bloom over Georges Bank and Nantucket Shoals
   Goes back years: Riley (1946) measured a strong spring (April) maximum in phytoplankton biomass over Georges Bank (between Cape Cod, Massachusetts, and Cape Sable Island, Nova Scotia) and a minimum in winter (January)

The bar graph represents the total phytoplankton concentration. All four regions show a clear seasonal cycle, and chlorophyll-a peaks typically occur in spring/early summer, with troughs in winter. It is clear that the data is primarily driven by diatoms, and by dinoflagellates, which form a majority of the total concentration. The 2021–2024 period has notably more outliers, and peak bloom magnitudes appear to have increased in the later period for Regions 1 and 2.

Bar graph outliers:

Region 1	Region 2	Region 3	Region 4
- March 2016 - September 2019 - March 2022 - July 2023 - October 2023 - November 2023	- February 2019 - March 2023 - November 2023 - March 2024	- January 2017 - December 2019 - July 2022 - June 2023	- Feb 2018 - March 2020 - March 2023 - October 2024




Phytoplankton Chlorophyll-a Concentration
Diatoms
Dinoflagellates
Green Algae
Prymnesiophytes
Other
No data (cloud cover)
Outlier
Region 1
Region 2
Region 3
Region 4
Chlorophyll-A Concentration (mg per m^3)
6.2
4.7
3.1
1.6
0.0
12.9
9.7
6.5
3.2
0.0
9.7
7.2
4.8
2.4
0.0
5.9
4.4
2.9
1.5
0.0
Apr
Aug
2021
2022
2023
2024

Outliers

Region	Date	Diatom peak	Dinoflagellate peak	Nearby storm	Nearby heatwave	Likely driver
Region 1	Jan-Feb 2016	✓ (Jan)	✓ (Feb)	Jan 2016 US blizzard		Storm
	Oct 2019	✓	✓	Oct 2019 bomb cyclone		Storm
	Jun 2023	✓	✓		2023 Gulf of Maine MHWs	Heatwave
	Jul 2023	✓			2023 Gulf of Maine MHWs	Heatwave
	Sep 2023		✓	Sep 2023 floods	2023 Gulf of Maine MHWs	Storm + heatwave
	Oct 2023	✓	✓		2023 Gulf of Maine MHWs	Heatwave
Region 2	Jun 2023	✓	✓		2023 Gulf of Maine MHWs	Heatwav
	Nov 2023	✓	✓		2023 Gulf of Maine MHWs (lingering)	Heatwave
	Dec 2024	✓	✓	Dec 2024 atmospheric river		Storm
Region 3	Jan-Feb 2017	✓ (Jan)	✓ (Feb)		Early-2017 Mid-Atlantic Bight advective heatwave	Heatwave
Region 3	Nov-Dec 2019	✓ (Dec)	✓ (Nov)			Unclear (storm/runoff/artifact)
Region 3	Jul 2022	✓	✓			Unclear

Region 4	Feb-Mar 2018	✓ (Mar)	✓ (Feb)	Mar 2018 Nor'easter		Storm
Region 4	Apr-Jun 2020	✓ (Jun)	✓ (Apr)			Unclear

Outliers Across Regions and Their Relationship to Storm Data

Diatom concentration peaks	Dinoflagellate concentration peaks	Storm data
• Region 1: January 2016, October 2019, June, July & October 2023 • Region 2: June & November 2023, December 2024 • Region 3: January 2017, December 2019, July 2022 • Region 4: March 2018, June 2020	• Region 1: February 2016, October 2019, June & September & October 2023 • Region 2: June & November 2023, December 2024 • Region 3: February 2017, November 2019, July 2022 • Region 4: February 2018, April 2020 Major storm events	• January 2016 United States blizzard • March 2018 Nor'easter • October 2019 Autumn storm "bomb cyclone" * September 2023 floods • December 2024 – Atmospheric river–fueled storm

Several of the biggest bloom outliers line up directly with Massachusetts storm events. Storms mix nutrient-rich deep water up to the surface and wash extra nutrients off land, giving both diatoms and dinoflagellates a feeding opportunity at the same time. Storm-driven turbulence can also resuspend dormant phytoplankton resting in the sediment, triggering new bloom events.

Shi, H., Chen, Y., Zhao, H., Mortimer, R., & Pan, G. (2025). Impact of tropical cyclone on coastal phytoplankton blooms and underlying mechanisms. Journal of Hydrology: Regional Studies, 59, 102389. https://doi.org/10.1016/j.ejrh.2025.10238

Outliers Across Regions and Their Relationship to Marine Heatwaves

The chlorophyll-a data reveals distinct outlier spikes across all four regions between 2016 and 2024. Several of these align closely with well-documented marine heatwave events, while others appear to be driven by more localized factors.

2016–2017, Clear match: Region 1's March 2016 outlier and Region 3's January 2017 outlier align with the 2016 Northwest Atlantic marine heatwave and the early-2017 advective heatwave in the Middle Atlantic Bight.

2018–2020, No clean match: Outliers in Region 4 (Mar 2018, Mar 2020), Region 2 (Feb 2019), and Region 3 (Dec 2019) don't correspond to a heatwave. These years did see notable storm activity, so runoff, storms, or cloud-cover artifacts are more likely explanations than heatwaves.

2021–2022, No flagged outliers, but notable warmth: No outliers appear, however these were the two warmest years on record for the Gulf of Maine. This may be because the warming was gradual rather than spiking sharply.

2023, Strongest cluster: Outliers appear across all four regions (Region 1: Jun/Oct, Region 2: Jun, Region 3: Jun, Region 4: Mar), matching the Gulf of Maine's three separate marine heatwaves that year.

Late 2023–2024, Uncertain: Outliers in Nov 2023 (Regions 1, 2), Dec 2024 (Region 2), and Oct 2024 (Region 4) may be lingering effects from the 2023 heatwaves or newer, undocumented warm spells.

Overall, the 2016, 2017, and 2023 outliers track known heatwaves well. The 2018 to 2020 outliers more likely reflect storms or data artifacts than basin-scale warming.

Open Ended Questions

1. How are the four regions different, and how does that change the data?
2. What would you want to know to decide if this pattern is a one-time anomaly or a new normal?
3. If you had to guess one environmental factor causing the outlier years, what would you look at first?
4. Why is knowing the specific species concentrations on top of total concentration important?
5. Why might a single-month spike be very different from a whole elevated year?

Other Research

Q. Recent research has shown that even Earth's most abundant photosynthetic microorganisms have environmental limits. If climate change alters which phytoplankton groups dominate our oceans, how might this reshape marine food webs, carbon storage, and the oxygen that supports life on Earth?

This study showed that **Prochlorococcus**, the most abundant photosynthetic organism on Earth and a major contributor to global oxygen production, has **thermal limits**. While it thrives in warm oceans, prolonged or extreme ocean warming can push it beyond its optimal temperature range, reducing growth and survival.

This challenges the common assumption that abundant marine microorganisms will simply adapt indefinitely to climate change.

Although **Prochlorococcus is not one of the four groups we analyzed**, the paper reinforces the broader scientific idea that:

- Different phytoplankton groups have different environmental tolerances,
- Warming oceans may change **which groups dominate**, not just the total amount of phytoplankton,
- Shifts in phytoplankton communities could alter marine food webs, oxygen production, and carbon cycling.

References:

1. Ribalet, F., Dutkiewicz, S., Monier, E., & Armbrust, E. V. (2025). *Future ocean warming may cause large reductions in Prochlorococcus biomass and productivity*. *Nature Microbiology*. Advance online publication. https://doi.org/10.1038/s41564-025-02106-4
2. National Oceanic and Atmospheric Administration. (n.d.). *How much oxygen comes from the ocean?* NOAA Ocean Service. https://oceanservice.noaa.gov/facts/ocean-oxygen.html
3. National Oceanic and Atmospheric Administration. (n.d.). *What are phytoplankton?* NOAA Ocean Service. https://oceanservice.noaa.gov/facts/phyto.html
4. National Oceanic and Atmospheric Administration. (n.d.). *Marine pollution.* NOAA Ocean Service. https://oceanservice.noaa.gov/facts/pollution.html
5. National Oceanic and Atmospheric Administration. (n.d.). *Ocean color and chlorophyll-a.* NOAA CoastWatch. https://ecowatch.noaa.gov/thematic/chlorophyll-a
6. National Oceanic and Atmospheric Administration Fisheries. (n.d.). *Phytoplankton in the Northeast U.S. Shelf Ecosystem.* https://www.fisheries.noaa.gov/new-england-mid-atlantic/ecosystems/phytoplankton-northeast-us-shelf-ecosystem
7. United Nations. (n.d.). *The ocean and climate change.* https://www.un.org/en/climatechange/science/climate-issues/ocean
8. United Nations. (n.d.). *Life below water (Sustainable Development Goal 14).* https://www.un.org/sustainabledevelopment/oceans/